

Dataset quality is high at 99.8, but the current configuration is insufficient for classifier training.

The dataset contains only 38 images, which is too small for effective classifier training. While image quality is high at 99.7, the metadata table suffers from critical mapping issues, including 3 missing files and 3 label disagreements. Furthermore, the provided data is primarily tabular metadata rather than the required image pixel tensors. Although cleaning improved the table quality score from 85.1 to 100.0, the structural mismatches between the annotations and files remain a significant barrier.

Data quality before	After cleaning	Items	Numbers fact-checked
92.4/100	99.8/100	76 → 75	7

Cross-type findings

CRITICAL Table rows point to missing images

3 table rows reference image files that are not present in the dataset.

Recommendation: Remove these 3 rows from the table or locate the missing image files to ensure data integrity.

Evidence: mix_links_001

WARNING Unreferenced image files found

There are 4 image files in the dataset that have no corresponding row in the annotations table.

Recommendation: Update the annotations table to include these 4 unreferenced images or remove the files from the dataset.

Evidence: mix_links_001

CRITICAL Label disagreements between table and folders

The label column in the table disagrees with the file's folder location for 3 rows.

Recommendation: Manually audit these 3 rows to determine the correct label and reconcile the table with the folder structure.

Evidence: mix_links_001

CRITICAL Dataset quality limited by table-image mismatch

While image quality is high at 99.7 out of 100, the overall dataset utility is constrained by the table-to-file mapping issues where only 35 of 38 rows successfully match a file.

Recommendation: Prioritize resolving the mapping discrepancies to ensure the high-quality images are correctly labeled for training.

Evidence: mix_overview_001, mix_links_001

WARNING Duplicate row references to a single file

The evidence notes that 1 file is referenced by more than one row in the annotations table.

Recommendation: Inspect and remove duplicate table rows pointing to the same image file to prevent training bias.

Evidence: mix_links_001

Each data type

Each type was profiled, cleaned, analyzed, and reviewed by its own crew. Their full reports are in `cleaned_by_type.zip`.

Table · 1 file(s) · quality 85.1 → 100.0 · rows 38 → 37

- Data quality score improved: The raw data quality score started at 85.1 out of 100 and improved to 100.0 out of 100 after cleaning.
- Duplicate rows removed: The raw dataset contained 1 exact duplicate row, representing 2.63% of the data, which was successfully removed.
- Notes column missingness and imputation: The notes column had 44.74% missing values and 5 hidden-missing tokens in the raw dataset, and mode imputation significantly inflated the frequency of the most common category from 9 to 25.
- Confidence and area metrics skewness: Confidence has a skew of -0.30 and area_px has a skew of -0.21 in the cleaned dataset.
- Dataset format mismatch with user goal: The user goal is to train an image classifier, but the dataset provided is tabular metadata containing features rather than actual image pixel tensors or binary image data.

Image · 38 file(s) · quality 99.7 → 99.7 · images 38 → 38

- Extremely Small Total Dataset Size: The dataset contains only 38 images in total across all classes.
- Mild Class Imbalance Present: Circles represent 16 images (42.11%), squares 12 images (31.58%), and triangles 10 images (26.32%), yielding an imbalance ratio of 1.6 between the largest and smallest classes.
- Zero Corrupt Images Found: Out of 38 analyzed images, there are 0 corrupt files and 0 images needing EXIF rotation.
- No Duplicates Detected: There are 0 exact duplicate groups, 0 near-duplicate groups, and 0 cross-class duplicate files.
- High Initial Image Quality Score: The overall image data quality score is 99.7 out of 100, reflecting 0 deductions for low quality, corruption, or duplicates.
- No Suspected Mislabeling in Vision Review: Based on an AI vision review observation, there are 0 suspected mislabeled images across all classes.
- Uniform Image Dimensions and Formats: All 38 images maintain a uniform width and height of 128 pixels, utilize the RGB mode, and belong to the JPEG format.
- Cleaning Summary Shows No Image Loss: The cleaning pipeline processed 38 images before and 38 images after, resulting in 0 removed files while preserving a 99.7 out of 100 quality score.

Table-to-file links

annotations.csv, column 'image': 35 of 38 rows match a file; 3 point to missing files; 4 of 38 files have no row; column 'label' disagrees with the folder for 3 rows.

Problem	Example
Row names a missing file	circle_100.jpg
Row names a missing file	circle_101.jpg
Row names a missing file	circle_102.jpg
File with no row	shapes_survey/images/circles/circle_003.jpg
File with no row	shapes_survey/images/squares/square_012.jpg
File with no row	shapes_survey/images/triangles/triangle_005.jpg
File with no row	shapes_survey/images/triangles/triangle_010.jpg
Label disagrees (row 4)	shapes_survey/images/circles/circle_004.jpg: table 'square', folder 'circles'
Label disagrees (row 14)	shapes_survey/images/circles/circle_014.jpg: table 'triangle', folder 'circles'
Label disagrees (row 24)	shapes_survey/images/squares/square_008.jpg: table 'circle', folder 'squares'

Next steps

1. Resolve the 3 critical label disagreements between the table and folder structure.
2. Address the 3 missing file references to ensure 1:1 mapping between annotations and images.
3. Verify the utility of the 'notes' column given that mode imputation increased the frequency of the common category from 9 to 25.
4. Collect additional data to mitigate the 1.6 class imbalance ratio and increase the total sample size beyond 38 images.
5. Convert the existing tabular metadata into a format compatible with standard image classification pipelines.

How the agents checked themselves

Model calls 27 · guardrail catches 7 · self-corrections 8 · review rounds 6 · model switches 7

+6s Switched models for the strategist

gemini-3.8-flash hit its daily limit; trying the next model in the pool.

+7s Switched models for the strategist

gemini-3.7-flash hit its daily limit; trying the next model in the pool.

+9s Switched models for the strategist

gemini-3.7-flash hit its daily limit; trying the next model in the pool.

+9s Switched models for the strategist

gemini-3.6-flash hit its daily limit; trying the next model in the pool.

+11s Switched models for the strategist

gemini-3.6-flash hit its daily limit; trying the next model in the pool.

+15s Switched models for the strategist

gemini-3.5-flash hit its server error 503 (set aside for 30s); trying the next model in the pool.

+15s Switched models for the strategist

gemini-3.5-flash hit its server error 503 (set aside for 120s); trying the next model in the pool.

+18s Cleaning plan failed the dry run

Step 6 (impute_median on ['confidence', 'area_px']) failed: TypeError: Cannot perform reduction 'median' with string dtype. The column is still text: convert it first (cast_numeric, strip_currency, or parse_percent).

+21s Revised output accepted

+24s Findings failed the fact check

Finding 7 ('Uniform Image Dimensions and Formats'): 'widths.min' isn't in ['img_overview_001', 'img_overview_002']. Use '<column>.<stat>' keys that appear in the evidence, and cite the artifact that contains them. Finding 7 ('Uniform Image Dimensions and Formats'): 'widths.max' isn't in ['img_ove...

+26s Findings failed the fact check

Finding 4 ('Class distribution in target labels'): 'label.top.circle' isn't in ['tbl_columns_001']. Use '<column>.<stat>' keys that appear in the evidence, and cite the artifact that contains them. Finding 4 ('Class distribution in target labels'): 'label.top.square' isn't in ['tbl_columns_001']....

+28s Revised output accepted

+28s Revised output accepted

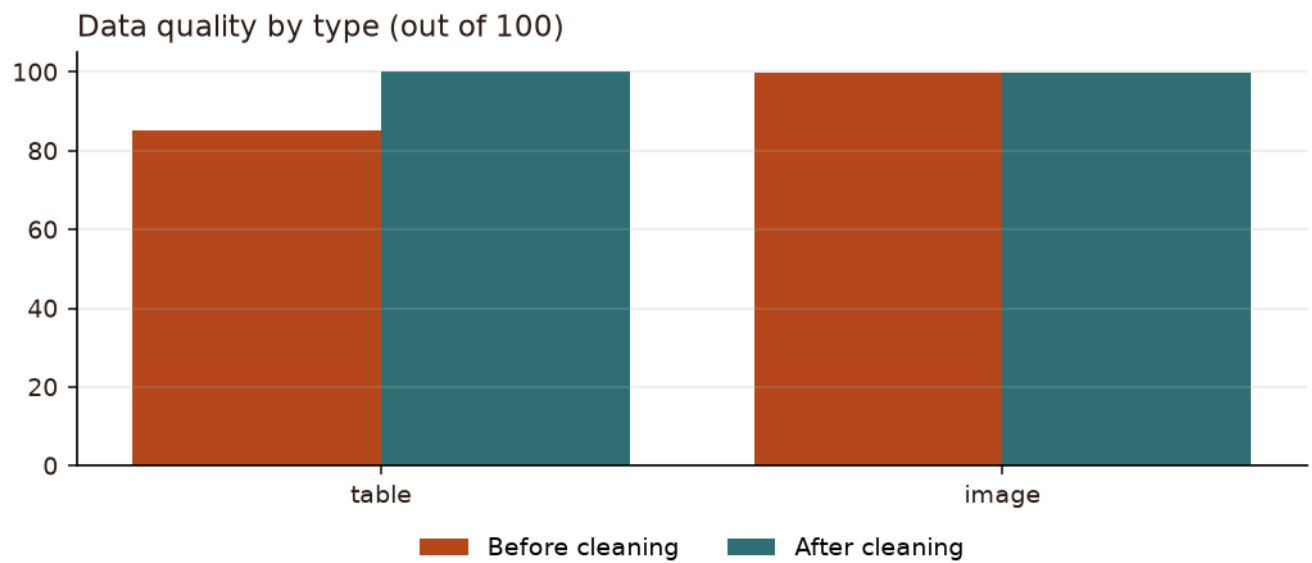
+31s Reviewer requested 1 change(s)

Finding 1 (severity): Finding 1 is marked as critical, but 38 images will not break a model or cause data leakage; small datasets are a warning/limitation rather than a critical model-breaking failure. Fix: Downgrade finding 1 from critical to warning.
... and 11 more in the HTML report.

Cleaning applied

Each data type was cleaned by its own crew.

Charts



Every number in the findings was checked against the evidence artifacts listed. Models used: gemini-3.1-flash-lite, gemini-3.5-flash-lite.